Centered-encoding parametrization for the MMS sweep

Working note (not yet merged into `article.tex`)

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1 Motivation

The stabilized FE solver in this codebase is dimensional. Its inputs are $(\rho, \mu, \sigma, U, L, \boldsymbol{f})$ in a chosen system of units. The Method of Manufactured Solutions (MMS) test harness, however, organises runs by the *dimensionless* regime (Re, Da, α_0), and is free to choose any consistent dimensional encoding of each dimensionless cell. Here α_0 is the infimum of the porosity field $\alpha(x)$, and $\alpha_\infty = 1$ is its supremum (a convention shared with `article.tex`).

Until recently, the harness fixed

$$U = 1, \qquad L = 1, \tag{1}$$

and derived ν, σ from the dimensionless definitions used in `article.tex`:

$$\nu \;=\; \frac{U\,L}{Re} \;=\; \frac{1}{Re}, \qquad \sigma \;=\; \frac{Da\;\alpha_\infty\,\nu}{L^2} \;=\; \frac{Da\;\alpha_\infty}{Re}. \tag{2}$$

For the sweep grid $Re \in \{10^{-6}, 1, 10^{6}\}$, $Da \in \{10^{-6}, 1, 10^{6}\}$, $\alpha_0 \in \{1, 0.5, 0.05\}$ (with $\alpha_\infty = 1$), this encoding produces values such as $\nu = 10^6$ and $\sigma = 10^{12}$ for the cell $(Re = 10^{-6}, Da = 10^6, \alpha_0 = 1)$. Twelve orders of magnitude separate the smallest from the largest dimensional coefficient that the solver sees, *all on the same side of unity.*

While the dimensionless physics is unchanged, this one-sided spread breaks numerical components of the stabilized solver that are not scale-invariant once round-off enters:

- the OSGS staggered fixed-point's L^2 projection π_h of the strong residual evaluates $\sigma\,\boldsymbol{u}_h$ and $\nu\Delta\boldsymbol{u}_h$ at near-precision-floor $\boldsymbol{u}_h$ when $\boldsymbol{u}_h \sim 1/\sigma$, amplifying floating-point noise into a non-contractive π_h update sequence;
- the assembled Jacobian's row magnitudes span 12 orders, which the direct factorisation absorbs but the Newton merit function and line-search tolerances do not;
- the MMS forcing oracle, computed by symbolic differentiation of the analytic exact solution, mixes $\sigma\,\boldsymbol{u}_{\mathrm{ex}}$ at magnitude 10^{12} with sub-dominant terms at 10^6, losing six digits of relative precision in the sub-dominant contributions.

2 The centered encoding

For a fixed dimensionless cell (Re, Da, α_∞), the harness has *two* free dimensional parameters (U, L). The freedom follows from the constraint pair (matching `article.tex` eq. 3.4):

$$Re = \frac{U\,L}{\nu}, \qquad\qquad Da = \frac{\sigma\,L^2}{\alpha_\infty\,\nu}. \tag{3}$$

Eliminating ν, σ:

$$\nu\,\sigma \;=\; \frac{\alpha_\infty\ Da\ U^2}{Re^2}, \qquad \frac{\nu}{\sigma} \;=\; \frac{L^2}{\alpha_\infty\ Da}. \tag{4}$$

So U controls the geometric mean and L controls the spread.

Choice. The *centered encoding* chooses U so that $\sqrt{\nu\,\sigma} = 1$, leaving $L = 1$ because the manufactured solution shape $\sin(\pi x)$ is hard-coded for a unit domain (see `Paper2DMMS` in `src/problems/mms_paper_2d.jl`):

$$\boxed{U \;=\; \frac{Re}{\sqrt{\alpha_\infty\ Da}}, \qquad L \;=\; 1.} \tag{5}$$

The resulting dimensional coefficients are

$$\nu \;=\; \frac{1}{\sqrt{\alpha_\infty\ Da}}, \qquad \sigma \;=\; \sqrt{\alpha_\infty\ Da}. \tag{6}$$

One checks immediately that $\nu\,\sigma = 1$ (centering condition) and $\nu/\sigma = 1/(\alpha_\infty\ Da)$ (intrinsic spread). Note that ν and σ no longer depend on Re — only U absorbs the Re variation. The spread $\sigma/\nu = \alpha_\infty\ Da$ is *intrinsic* to the cell and cannot be compressed by any choice of (U, L); centering only shifts it to be symmetric about unity.

What this does *not* change. The dimensionless equation (and therefore the dimensionless solution $\boldsymbol{u}^*, p^*$) is identical between the default and centered encodings; this is a strict reparametrization. The discrete velocity *degrees of freedom* are $\boldsymbol{u}_h^{\mathrm{node}} \approx U\,\boldsymbol{u}^{*,\mathrm{node}}$, so they take on different absolute magnitudes between encodings, but their dimensionless counterparts $\boldsymbol{u}^{*,\mathrm{node}}$ are unchanged. The H^1-seminorm error $|\boldsymbol{u}_h - \boldsymbol{u}_{\mathrm{ex}}|_{H^1(\Omega)}\,/\,(U/L)$ is regime-invariant (verified empirically: agreement to $<1\%$ across encodings on the easy regime C2).

3 Per-cell parameter table

The harness sweeps $Re \in \{10^{-6}, 1, 10^{6}\}$, $Da \in \{10^{-6}, 1, 10^{6}\}$, $\alpha_0 \in \{1, 0.5, 0.05\}$, with $\alpha_\infty = 1$ in all cells. Three cells are skipped by config ($(Re, Da, \alpha_0) = (10^6, \star, 0.05)$, three triples), leaving 24 that are exercised by the sweep. Table 1 reports the dimensional values for both encodings. Because the dimensional σ in the code path (`sigma_c = Da * alpha_infty * nu_calc / L^2`) uses $\alpha_\infty = 1$ throughout, the σ columns are independent of α_0 — they are determined purely by (Re, Da).

Maximum dimensional coefficient. The worst cells under the default encoding (C7, C16, C23) hit $\sigma = 10^{12}$. Under the centered encoding the worst is $\sigma = 10^3$ (any cell with $Da = 10^6$). The maximum dimensional coefficient drops by nine orders of magnitude across the sweep.

4 Verification evidence

Two single-cell probes (`test/extended/ManufacturedSolutions/diagnostics/jacobian_equilibration_osgs_` validated the reparametrization before the full rerun. Note that the probe used a slightly different centering formula $U = Re\,/\sqrt{\alpha_0\,Da}$ (with $\alpha_0 = 0.5$ for C16), so $U = 1.41 \cdot 10^{-9}$ and $\sigma = 1.41 \cdot 10^{3}$ in the probe; the harness rerun uses $U = Re\,/\sqrt{\alpha_\infty\,Da}$ with $\alpha_\infty = 1$, giving $U = 10^{-9}$ and $\sigma = 10^3$ exactly.

#	Dimensionless inputs			Default ($U = L = 1$)			Centered		
	Re	Da	α_0	U	ν	σ	U	ν	σ
C1	10^{-6}	10^{-6}	1.00	1	10^{6}	1	10^{-3}	10^{3}	10^{-3}
C2	1	10^{-6}	1.00	1	1	10^{-6}	10^{3}	10^{3}	10^{-3}
C3	10^{6}	10^{-6}	1.00	1	10^{-6}	10^{-12}	10^{9}	10^{3}	10^{-3}
C4	10^{-6}	1	1.00	1	10^{6}	10^{6}	10^{-6}	1	1
C5	1	1	1.00	1	1	1	1	1	1
C6	10^{6}	1	1.00	1	10^{-6}	10^{-6}	10^{6}	1	1
C7	10^{-6}	10^{6}	1.00	1	10^{6}	10^{12}	10^{-9}	10^{-3}	10^{3}
C8	1	10^{6}	1.00	1	1	10^{6}	10^{-3}	10^{-3}	10^{3}
C9	10^{6}	10^{6}	1.00	1	10^{-6}	1	10^{3}	10^{-3}	10^{3}
C10	10^{-6}	10^{-6}	0.50	1	10^{6}	1	10^{-3}	10^{3}	10^{-3}
C11	1	10^{-6}	0.50	1	1	10^{-6}	10^{3}	10^{3}	10^{-3}
C12	10^{6}	10^{-6}	0.50	1	10^{-6}	10^{-12}	10^{9}	10^{3}	10^{-3}
C13	10^{-6}	1	0.50	1	10^{6}	10^{6}	10^{-6}	1	1
C14	1	1	0.50	1	1	1	1	1	1
C15	10^{6}	1	0.50	1	10^{-6}	10^{-6}	10^{6}	1	1
C16	10^{-6}	10^{6}	0.50	1	10^{6}	10^{12}	10^{-9}	10^{-3}	10^{3}
C17	1	10^{6}	0.50	1	1	10^{6}	10^{-3}	10^{-3}	10^{3}
C18	10^{6}	10^{6}	0.50	1	10^{-6}	1	10^{3}	10^{-3}	10^{3}
C19	10^{-6}	10^{-6}	0.05	1	10^{6}	1	10^{-3}	10^{3}	10^{-3}
C20	1	10^{-6}	0.05	1	1	10^{-6}	10^{3}	10^{3}	10^{-3}
(skipped: $Re = 10^6, Da = 10^{-6}, \alpha_0 = 0.05$)									
C21	10^{-6}	1	0.05	1	10^{6}	10^{6}	10^{-6}	1	1
C22	1	1	0.05	1	1	1	1	1	1
(skipped: $Re = 10^6, Da = 1, \alpha_0 = 0.05$)									
C23	10^{-6}	10^{6}	0.05	1	10^{6}	10^{12}	10^{-9}	10^{-3}	10^{3}
C24	1	10^{6}	0.05	1	1	10^{6}	10^{-3}	10^{-3}	10^{3}
(skipped: $Re = 10^6, Da = 10^6, \alpha_0 = 0.05$)									

Table 1: Dimensional parameter values for each dimensionless cell in the `phase1_quad_k1` sweep, under the default encoding ($U = L = 1$) and the centered encoding ($U = Re/\sqrt{\alpha_\infty Da}$, $L = 1$), with $\alpha_\infty = 1$ as run by the test harness (`run_test.jl:76`). The dimensionless physics is identical between the two columns; only the dimensional representation differs. Under the centered encoding, ν and σ depend only on Da, never on Re or α_0.

C16 OSGS (high-σ, the canonical fold case).

	Default	Centered (probe)
ν	10^6	$1.41 \cdot 10^{-3}$
σ	10^{12}	$1.41 \cdot 10^3$
OSGS staggered iters	25 (budget)	0 (immediate)
π_u-drift	$7.6 \cdot 10^8$	$1.03 \cdot 10^{-6}$
`overall_converged`	`false`	`true`
Initial residual	$1.27 \cdot 10^{-3}$	$3.08 \cdot 10^{-10}$
$L^2(\boldsymbol{u})$	$1.07 \cdot 10^{-3}$	$4.72 \cdot 10^{-4}$
$H^1(\boldsymbol{u})$	$2.45 \cdot 10^{-1}$	$1.04 \cdot 10^{-1}$

The OSGS staggered fixed-point's projection drift drops by *fourteen orders of magnitude*. Same physics; the difference is the floating-point conditioning of the strong-residual projection.

C2 (easy regime; consistency check). With $Re = 1$, $Da = 10^{-6}$, $\alpha_0 = 1$, both encodings should give the same answer up to round-off.

	Default ($U = 1$)	Centered ($U = 10^3$)	
$H^1(\boldsymbol{u})$	$3.612 \cdot 10^{-2}$	$3.564 \cdot 10^{-2}$	(0.7% diff)
$L^2(\boldsymbol{u})$	$5.92 \cdot 10^{-4}$	$1.18 \cdot 10^{-4}$	(different noise floor)

The H^1 semi-norm error agrees to under 1% between encodings, confirming that the centered encoding represents the same discrete physics. The L^2 values differ by a constant noise-floor factor: in both runs the absolute residual at convergence is $O(10^{-9})$, and that absolute noise is divided by $U_c = U$ when forming the normalized error — a 1000× different denominator produces the apparent 5× discrepancy without any discretisation disagreement.

5 Implementation

The implementation is a single-line change to the MMS harness; no modification to the dimensional solver code:

```
# test/extended/ManufacturedSolutions/run_test.jl, ~line 523

# Before (default encoding):
U_amp = 1.0
L     = 1.0

# After (centered encoding):
U_amp = Re / sqrt(alpha_infty * Da)
L     = 1.0
```

The remainder of the harness pipeline (`build_mms_formulation`, `Paper2DMMS`, `evaluate_exactness_diagnosti` `get_characteristic_scales`) already threads U and L through correctly: the MMS exact velocity is multiplied by U, the manufactured forcing is computed analytically from that scaled solution, and error norms divide out $U_c = U$ to report dimensionless quantities. No tooling surgery is required.

6 Exploring the second degree of freedom: variable L

The centered encoding of Section 2 spends the U degree of freedom and hardcodes $L = 1$ because the manufactured shape $\sin(\pi x)$ is wired to a unit period. With a polynomial generalisation $\sin(\pi x/L)$ and a domain that scales with L (so $h \mapsto L/N$ and the dimensionless analysis remains intact), L becomes free as well and the dimensional spread can be compressed further. This section quantifies how much.

6.1 Identities and the design space

Recall the two identities from Section 2:

$$\nu\,\sigma \;=\; \frac{\alpha_\infty\; Da\; U^2}{Re^2}, \qquad \frac{\nu}{\sigma} \;=\; \frac{L^2}{\alpha_\infty\; Da}. \tag{7}$$

The first identity says U alone controls the geometric mean $\sqrt{\nu\sigma}$ — L has no effect on it. The second says L alone controls the ratio ν/σ — U has no effect on it. The two degrees of freedom are therefore *orthogonal* in log-space: one slides the pair (ν, σ) along the line $\log\nu + \log\sigma = \text{const}$, the other rotates the pair around its geometric mean.

6.2 Balanced-coefficient encoding ($\nu = \sigma$)

The natural use of the L degree of freedom is to set $\nu/\sigma = 1$, making the diffusion and reaction coefficients dimensionally identical on every cell. From the second identity in (7):

$$L^\star \;=\; \sqrt{\alpha_\infty\; Da}. \tag{8}$$

Note that, like ν and σ in the centered encoding, $L^\star$ depends only on Da (and α_∞), not on Re or α_0.

The remaining choice is U. The centered selection $U = Re\,/\sqrt{\alpha_\infty\; Da}$ collapses the geometric mean to unity ($\sqrt{\nu\sigma} = 1$), so combined with (8) it yields $\nu = \sigma = 1$ on every cell. This is the most aggressive flattening of the operator coefficients, but it leaves U unchanged from the centered encoding — the same factor of 10^9 excursion to $U = 10^{-9}$ at the C7 corner.

6.3 Minmax encoding: trading U headroom against ν, σ

A different use of the L DOF is to minimise the worst-case excursion across all three of U, ν, σ simultaneously. Concretely, we solve

$$\min_{U,L}\;\; \max\bigl(\,|\log_{10} U|,\; |\log_{10}\nu|,\; |\log_{10}\sigma|\,\bigr). \tag{9}$$

Keeping $\nu = \sigma$ (which sets $L = L^\star$ as above) and choosing U to equalise $|\log U|$ with $|\log\nu|$ gives a closed form. Define

$$A \;\equiv\; \tfrac{1}{2}\log_{10}(\alpha_\infty\, Da) \;-\; \log_{10} Re\,. \tag{10}$$

A is the excursion of $\log\nu$ from $\log U$ at the centered choice. The minmax optimum is achieved at

$$\boxed{\log_{10} U^{\star\star} \;=\; -\frac{A}{2}, \qquad \log_{10} L^{\star\star} \;=\; \tfrac{1}{2}\log_{10}(\alpha_\infty\, Da), \qquad \log_{10}\nu^{\star\star} \;=\; \log_{10}\sigma^{\star\star} \;=\; +\frac{A}{2}.} \tag{11}$$

At this point, $|\log U| = |\log \nu| = |\log \sigma| = |A|/2$ — all three dimensional quantities sit at the same distance from unity, with U on the opposite side of σ in log-space. Closed forms:

$$U^{\star\star} = \left(\frac{Re^2}{\alpha_\infty\ Da}\right)^{1/4}, \qquad L^{\star\star} = \sqrt{\alpha_\infty\ Da}, \qquad \nu^{\star\star} = \sigma^{\star\star} = \left(\frac{\alpha_\infty\ Da}{Re^2}\right)^{1/4}. \tag{12}$$

Notice the satisfying symmetry $U^{\star\star} \cdot \nu^{\star\star} = 1$ on every cell.

Spread reduction. Under the centered encoding the worst-case excursion is $|A|$: $\log \nu = -A/2 \cdot 2 =$ (signed) A and $\log U =$ opposite-sign $A/2$, so $\max(|\log U|, |\log \nu|, |\log \sigma|) = |A|$. Under minmax that worst case drops to $|A|/2$ — **a factor-of-two reduction in log-magnitude**, i.e. the worst dimensional quantity halves its number of digits away from unity, on every cell.

6.4 Per-cell numerical values

Table 2 reports the dimensional values under both new strategies, alongside the centered baseline for reference. All values shown for $\alpha_\infty = 1$ as run by the harness.

	Dim'less			Centered ($L=1$)			Balanced ($\nu=\sigma$, U centered)				Minmax ($\nu=\sigma=1/U$)			
#	Re	Da	α_0	U	ν	σ	U	L	ν	σ	U	L	ν	σ
C1	10^{-6}	10^{-6}	1.00	10^{-3}	10^{3}	10^{-3}	10^{-3}	10^{-3}	1	1	$10^{-1.5}$	10^{-3}	$10^{1.5}$	$10^{1.5}$
C2	1	10^{-6}	1.00	10^{3}	10^{3}	10^{-3}	10^{3}	10^{-3}	1	1	$10^{1.5}$	10^{-3}	$10^{-1.5}$	$10^{-1.5}$
C3	10^{6}	10^{-6}	1.00	10^{9}	10^{3}	10^{-3}	10^{9}	10^{-3}	1	1	$10^{4.5}$	10^{-3}	$10^{-4.5}$	$10^{-4.5}$
C4	10^{-6}	1	1.00	10^{-6}	1	1	10^{-6}	1	1	1	10^{-3}	1	10^{3}	10^{3}
C5	1	1	1.00	1	1	1	1	1	1	1	1	1	1	1
C6	10^{6}	1	1.00	10^{6}	1	1	10^{6}	1	1	1	10^{3}	1	10^{-3}	10^{-3}
C7	10^{-6}	10^{6}	1.00	10^{-9}	10^{-3}	10^{3}	10^{-9}	10^{3}	1	1	$10^{-4.5}$	10^{3}	$10^{4.5}$	$10^{4.5}$
C8	1	10^{6}	1.00	10^{-3}	10^{-3}	10^{3}	10^{-3}	10^{3}	1	1	$10^{-1.5}$	10^{3}	$10^{1.5}$	$10^{1.5}$
C9	10^{6}	10^{6}	1.00	10^{3}	10^{-3}	10^{3}	10^{3}	10^{3}	1	1	$10^{1.5}$	10^{3}	$10^{-1.5}$	$10^{-1.5}$
$\alpha_0 = 0.5$ *cells (C10–C18): identical* U, ν, σ, L *values as C1–C9 because the harness uses* $\alpha_\infty = 1$.														
C19	10^{-6}	10^{-6}	0.05	10^{-3}	10^{3}	10^{-3}	10^{-3}	10^{-3}	1	1	$10^{-1.5}$	10^{-3}	$10^{1.5}$	$10^{1.5}$
C20	1	10^{-6}	0.05	10^{3}	10^{3}	10^{-3}	10^{3}	10^{-3}	1	1	$10^{1.5}$	10^{-3}	$10^{-1.5}$	$10^{-1.5}$
C21	10^{-6}	1	0.05	10^{-6}	1	1	10^{-6}	1	1	1	10^{-3}	1	10^{3}	10^{3}
C22	1	1	0.05	1	1	1	1	1	1	1	1	1	1	1
C23	10^{-6}	10^{6}	0.05	10^{-9}	10^{-3}	10^{3}	10^{-9}	10^{3}	1	1	$10^{-4.5}$	10^{3}	$10^{4.5}$	$10^{4.5}$
C24	1	10^{6}	0.05	10^{-3}	10^{-3}	10^{3}	10^{-3}	10^{3}	1	1	$10^{-1.5}$	10^{3}	$10^{1.5}$	$10^{1.5}$

Table 2: Per-cell dimensional values under three encodings: centered ($L = 1$ fixed), balanced ($\nu = \sigma$ via $L = \sqrt{\alpha_\infty Da}$ with centered U), and minmax (jointly optimal U and L). Skipped cells ($Re = 10^6, \alpha_0 = 0.05$) omitted. Values shown with $\alpha_\infty = 1$ as run by the test harness; under this convention all three encodings are independent of α_0.

6.5 Comparison summary

For each encoding and each cell, the worst-case dimensional excursion $M = \max(|\log_{10} U|, |\log_{10} \nu|, |\log_{10} \sigma|)$ measures how far from unity the most extreme coefficient lies. The sweep maxima over all 24 cells are:

Encoding	$\max_{\text{cells}} M$
Default ($U = L = 1$)	12 (C7, C23)
Centered ($L = 1$)	9 (C3, C7, C9, C12, C16, C18, C23)
Balanced ($\nu = \sigma$)	9 (C3, C7, C9, C12, C23) — still set by U
Minmax (jointly optimal U, L)	4.5 (C3, C7, C9, C12, C23)

The balanced encoding flattens the operator coefficients but does not improve the worst-case excursion because the U value is unchanged from centered. The minmax encoding redistributes the excursion across all three quantities and halves the worst case from 9 to 4.5 orders of magnitude.

6.6 Whether this pays off in practice

The minmax encoding is mathematically attractive but *not without cost.* Two effects work against the reduced excursion:

1. **U becomes finite-but-small at high Da even after minmax.** For C7, $U^{\star\star} = 10^{-4.5} \approx 3.2 \times 10^{-5}$ compared to $U^{\text{centered}} = 10^{-9}$. A factor of $\sim 10^{4.5}$ improvement on solution-magnitude precision: the absolute FE truncation error at $N = 320$ moves from $\sim 10^{-14}$ (at machine epsilon for the inner product) to $\sim 10^{-9.5}$ (well above the L^2 noise floor). If the observed C7 stall at the finest mesh is genuinely solution-scale-driven, the minmax encoding should cure it.

2. **ν becomes large at high Da.** For C7, $\nu^{\star\star} = 10^{4.5} \approx 3.2 \times 10^4$, a factor of $\sim 10^{7.5}$ larger than $\nu^{\text{centered}} = 10^{-3}$. Diffusion-bilinear-form entries scale with ν, so the Jacobian magnitude grows by the same factor. This is asymptotically harmless (Float64 happily represents $10^{4.5}$), but the assembled-system condition number and the line-search merit-function dynamic range both shift; existing solver tolerances calibrated for the centered regime may need a re-check.

The balanced encoding ($\nu = \sigma = 1$ everywhere) sidesteps *both* of these concerns at the cost of no U improvement: a strict superset of the precision benefits already validated for centered in Section 4, with the additional invariant that the assembled operator's diffusion and reaction blocks are dimensionally identical. Its main appeal is conceptual cleanliness, not precision recovery.

The choice between balanced and minmax therefore depends on the diagnosis of which precision floor is binding. The unit-amp probe described elsewhere in this session is the cheapest test: it isolates the U-side floor by holding $L = 1$ and forcing $U = 1$, decoupling the solution-scale effect from the operator-scale effect. If the probe recovers C7's slope, minmax is strictly preferable; if not, the balanced encoding (or the centered encoding already in use) is sufficient and the implementation cost of L generalisation is not justified.

6.7 Implementation cost summary

Generalising the manufactured solution to allow variable L is a contained refactor of `src/problems/mms_paper_2d.j` plus a domain-scaling hook in the harness. The polynomial frequency factor π becomes π/L in five functions (`UExFunc`, `grad_u_ex`, `lap_u_ex`, `grad_div_u_ex`, `get_p_ex`, plus the pressure-gradient block in `evaluate_exactness_diagnostics`), each derivative order picking up a $1/L$. The domain bounding box scales as $L \cdot [-0.5, 0.5]^2$, the porosity radii rescale as $L \cdot r_i$, and the per-cell loop in `run_test.jl` must rebuild the mesh and FE spaces inside the per-cell branch instead of per-N. Half a day to a full day of careful work plus a regression test that confirms $L = 1$ reproduces the master HDF5 bit-identically and an MMS test at $L = 10$ confirms identical relative slopes. The main risk is the h-coupling: at $L \neq 1$, the mesh-size-dependent stabilisation τ must be confirmed to scale correctly, which it does by the $\mathrm{Re}_h = \mathrm{Re}/N$ invariance above but requires a sanity check.

7 What is deliberately *not* addressed here

- **The L degree of freedom — analysed in Section 6.** `Paper2DMMS` hard-codes the spatial shape $\sin(\pi x_1)\sin(\pi x_2)$ etc., which is one half-period on the unit square but high-frequency garbage on $L \neq 1$. Allowing L to vary requires generalising the shape to $\sin(\pi x/L)$, the analytic gradient/Laplacian/forcing terms, the domain bounding box, and the porosity radii — a contained refactor of `src/problems/mms_paper_2d.jl` plus a per-cell mesh rebuild in the harness. Section 6 works out two natural strategies (balanced $\nu = \sigma$, and minmax) and their per-cell numerics; the minmax strategy halves the worst-case dimensional excursion from 9 to 4.5 orders of magnitude. The implementation is deferred pending diagnosis of whether the C7-class slope drop is solution-scale (in which case minmax cures it) or operator/assembly-scale (in which case it does not).
- **The pressure penalty $\eta = \text{eps\_val}$.** The dimensional pressure amplitude P_c varies with the encoding, which means $\eta \cdot p^2$ has a different effective weight per cell. Empirically the discrete LBB stabilization is the primary gauge fix and η acts only as a safety net; the rerun will validate this in practice. If the sweep reveals an η-related regression, a regime-aware scaling $\eta_{\text{cell}} \propto P_c^{-2}$ is a focused follow-up.
- **Solver tolerances.** `ftol`, `xtol`, `stagnation_noise_floor` are absolute. Under the centered encoding the residual magnitudes shrink for low-*Re* cells; the existing `dynamic_ftol_spatial_safety_factor` machinery scales `ftol` with h^{k+1}, which empirically reaches sensible values in both encodings (the probes converged to high precision). If a regression appears, the dynamic-tolerance formula is the first place to look.

Status

This document accompanies a single-line code change at `test/extended/ManufacturedSolutions/run_test.jl:5`. It is kept in `theory/` as a standalone working note pending the full 24-cell sweep rerun on this encoding; once the encoding is shown to be *uniformly* desirable (no regression on the easy cells, demonstrable gains on the high-σ cells), the content here should be folded into `article.tex` as a remark in the verification section.